

You are given a stack containing integers in arbitrary order. Your task is to write a function that segregates odd and even numbers while maintaining their relative order.

- Odd numbers should appear at the top (in the same order as in the original stack).
- Even numbers should appear at the bottom (in the same order as in the original stack).
- You can only use stack operations: push(value), pop(), and isEmpty() (or equivalent in Python).
- You may use multiple auxiliary stacks but cannot use other data structures like arrays or lists.

| Input                              | Output                             | Explanation                                                                                                                                                                                              |
|------------------------------------|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12(top) → 7 → 5 → 8 → 11 → 14 → 3  | 7(top) → 5 → 11 → 3 → 12 → 8 → 14  | The odd numbers (7,5,11) are grouped together and appear at the top of the output stack, in the same order as the original stack.<br><br>The even numbers appear next, maintaining their original order. |
| 1(top) → 2 → 3 → 4 → 5 → 6 → 7 → 8 | 1(top) → 3 → 5 → 7 → 2 → 4 → 6 → 8 | Same reasoning as before                                                                                                                                                                                 |

| Java Template                                                                | Python Template                                            |
|------------------------------------------------------------------------------|------------------------------------------------------------|
| <pre>public Stack segregate(Stack original) {     // Write your code }</pre> | <pre>def segregate(original):     // Write your code</pre> |